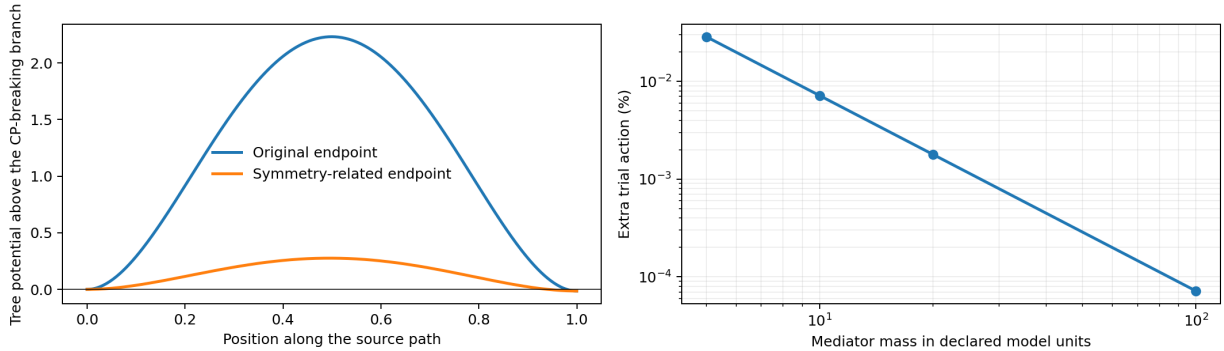


Flavor cosmology: metastable decay, CP walls and constrained bias

The current release turns the conditional wall interface into a calculation on the published flavor action. It consumes the lower CP-conserving competitor already found by the other flavor session, rather than searching for new stationary points or changing its coefficients. At the pinned main-branch commit 22e4edf06dbdeea0f9245dbb06669ed57b7906fc, the nominated CP-breaking branch has energy -4.196243686677212, while the retained stable CP-conserving competitor has energy -4.210439239580322. The gap is 0.01419555290310992. Both are locally stationary to approximately 10^{-15} in the declared numerical action, and their reported smallest physical Hessian eigenvalues are positive. The absolute global minimum remains unclassified. This benchmark therefore calls for a metastable-decay calculation, not a claim of stable walls between global CP-breaking vacua.

Published flavor action: trial decay paths and mediator kinetic cost



Tree-action trial profiles; no lifetime or physical nucleation rate is inferred.

The new reusable module is `python/perfectpower/flavor_decay.py`, with the driver `python/develop_flavor_cosmology.py` and eighteen focused tests in `python/tests/test_flavor_decay.py`. The scientific receipt is `receipts/flavor_cosmology/tree_decay.json`. It retains exact rational coefficients for the radial trial profiles, the input-file hashes, every chosen endpoint and the full 48-coordinate mediator-lifted path. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python python/develop_flavor_cosmology.py` from the repository root. No source couplings are retuned, no stationary points are re-minimized, and no quark matching is repeated. The earlier 104-check Node calculation remains an independent conditional interface.

The numerical source tensors, coefficients and endpoint coordinates are converted to exact binary fractions. Five exact evaluations along a source line reconstruct its quartic potential by rational Newton interpolation. This arithmetic is exact for the binary-rational realization of those arrays. It does not turn floating-point finite-group projectors into exact algebraic-number tensors, or approximate stationary coordinates into proven exact minima. Thirty-three independent ordinary floating-point evaluations per candidate check the reconstructed polynomial against the existing full potential. This distinction matters: the radial action integrals below are exact, but the physical identification with a certified bounce remains conditional on the false-vacuum and reduced-variational hypotheses.

Let $\gamma(q)$ be a polynomial path in canonically normalized real fields, with $\gamma(0)$ the supplied false vacuum and $\gamma(1)$ a lower-energy configuration. Let $W(q) = V(\gamma(q)) - V(\gamma(0))$. A compact $O(4)$ trial profile has $q=1$ inside a core, a smooth shell, and $q=0$ outside. Put $u=r/R$. The core is $0 \leq u \leq L$; in the shell write $t=u-L$ between zero and one and set $q(t) = 1 - 3t^2 + 2t^3$. This taper has zero derivative at both joins, so the profile is continuously differentiable. Outside $u=L+1$ the field equals the false background exactly. No exponential tail cutoff or divergent outside energy is introduced.

The complete action of this profile is

$$S_4(R, L) = 2\pi^2 [T(L) R^2 + U(L) R^4],$$

where $T(L) = \frac{1}{2}$ times the integral from zero to one of $(L+t)^3$ times the squared norm of $d\gamma(q(t))/dt$. The potential term is $U(L) = W(1)L^4/4$ plus the integral from zero to one of $(L+t)^3 W(q(t))$. Both are rational polynomials in L whenever the supplied field-path and potential coefficients are rational. T has degree at most three and U has degree at

most four. The core-volume contribution ensures $U < 0$ for sufficiently large L whenever $W(1) < 0$. This makes a negative-action escape configuration available without pretending the Euclidean bounce is a global action minimum.

For a retained shape with $T > 0$ and $U < 0$, the action as a function of size starts at zero, reaches a positive maximum and then becomes negative. The maximum lies at $R_{\text{crit}}^2 = -T/(2U)$, with $S_{\text{max}}/\pi^2 = -T^2/(2U)$. Its derivative under radial scaling vanishes there and its scaling second derivative is negative, as expected for the escape direction. The code evaluates these quantities as exact fractions. It uses a numerical one-parameter search only to choose a rational L and then recomputes the retained action exactly. It does not certify global optimality even within the shape family.

The appropriate variational interpretation is an upper estimate for the least decay barrier, under the usual reduced-problem or mountain-pass existence assumptions and an exact metastable exterior. Every such trial profile produces a route from the false configuration to negative action; the least separating saddle cannot have a larger minimax level than that route. This is not a lower bound on the bounce action. Consequently a large trial action cannot establish a long lifetime. The familiar scale-reduced formulation originates in Coleman, Glaser and Martin, Action Minima among Solutions to a Class of Euclidean Scalar Field Equations, Communications in Mathematical Physics 58 (1978) 211-221, DOI 10.1007/BF01609421. The multifield symmetry and reduced-problem existence assumptions are treated by Blum, Honda, Sato, Takimoto and Tobioka, $O(N)$ Invariance of the Multi-Field Bounce, arXiv:1611.04570, <https://arxiv.org/abs/1611.04570>. The present radial-profile construction and its integration are an adapter to this established framework.

A significant practical reduction comes from correctly using the internal symmetry. The driver evaluates the 1080 existing family matrices on the already-published lower endpoint, together with both sector-sign transformations. This supplies 4320 symmetry-related endpoint images; it does not produce 4320 independently minimized vacua. The source coefficients remain identical. Four nearest images and the original endpoint are retained as candidate straight paths. A central group image can repeat the same real-adjoint endpoint, so the retained list is not claimed to classify distinct physical vacua. A farther endpoint might have a better potential barrier; this finite nearest-image selection is deliberately a bounded path search, not an exhaustive optimization.

With the original endpoint, the retained compact-profile trial action is approximately 2.5305352605 times 10^{10} . The best retained symmetry-related endpoint yields approximately 3.3396009270 times 10^7 , a reduction by a factor of 757.7358241. Its squared source distance is 3.210476017821027; the original endpoint's squared distance is 11.166145333203302. Its core-to-shell ratio L is 45.94751799. These are dimensionless trial actions for the specified tree potential. Their numerical size is not a cosmological lifetime, a measured decay rate, or a statement that this action is near the actual least bounce. The result shows why symmetry-aware field-path selection materially affects even a conservative decay calculation.

The mediator completion introduces a further geometric issue. The positive Gaussian square allows the scalar potential to match the source potential exactly at $S = -J(x)/M^2$, but this equality does not erase the mediator kinetic energy when x varies across a bubble. Following that trough gives the induced field-space metric

$$ds^2 = dx^T dx + dJ^T dJ/M^4.$$

The driver constructs all 27 mediator coordinates as exact quadratic polynomials along the source line, retaining the 21 source/Higgs coordinates. The complete trial bubble therefore uses 48 canonical real fields. Its potential along the path is unchanged, while its kinetic integral increases by one half the radial integral of the squared mediator derivatives. The positive-square contribution is evaluated independently at seventeen path points; it is negligible at floating-point reconstruction precision. Potential matching and kinetic matching are thus kept separate.

At any fixed path and retained shape, the completion changes the candidate action by the exact factor $[1 + \Delta T(L)/T_{\text{source}}(L)]^2$. The extra kinetic integral is nonnegative, so this factor cannot decrease the action of that particular trial profile. This is not a theorem that the true UV least bounce is greater than the EFT least bounce: UV paths can leave the trough, and the least path has not been solved. The statement concerns the explicitly retained trough-following profiles. Treating Gaussian potential matching alone as sufficient for tunneling would omit this positive geometric cost.

The mass scan keeps the EFT coefficients fixed and scales the current couplings in the existing Gaussian realization proportionally to M . Then $S = -J/M^2$ scales as $1/M$, and the induced kinetic correction scales as $1/M^2$. For masses 5, 10, 20, 100 in the stated model units, the separately shape-optimized trough-profile action ratios to the best source candidate are approximately 1.0002861286, 1.0000715283, 1.0000178818, and 1.0000007153. At $M=10$ the lifted action is approximately 3.3398398030 times 10^7 , an increase of 0.00715283 percent. This scan demonstrates decoupling within the same completed tree construction. It does not vary the mediator mass while incorrectly freezing its current normalization, and it includes no mediator-loop Coleman–Weinberg correction.

The CP interpretation also becomes sharper. Direct evaluation of the supplied generalized-CP partner reproduces the false-branch energy, and CP covariance is checked on the false point, chosen lower point and their midpoint. CP-even matching or subsequent CP-even thermal evolution cannot select a weak-CP sign merely by splitting this conjugate pair. If the CP-conserving lower basin ultimately occupies the universe, it removes the physical weak-CP order parameter instead of predicting its observed sign. Whether this basin is populated requires nucleation dynamics, initial conditions and an adequately controlled action; the local CP-breaking benchmark alone supplies none of those.

The upstream quantum calculation is a substantial limit on physical interpretation. At its fixed $\mu=1$ inputs, the linearized one-loop vacuum displacement is 15.51581387 times the tree-background norm. This signals an uncontrolled small-displacement expansion. A tree trial action cannot be presented as the quantum-corrected tunneling exponent for that benchmark. The current result deliberately imports and displays that warning alongside its own action rather than fitting counterterms or changing couplings to conceal it. A controlled quantum action is an input the mechanism session must eventually supply before physical decay claims become meaningful.

The upstream leading high-temperature minimal-action Hessian is positive, supporting asymptotic restoration within its declared expansion. That does not provide a transition temperature or an $O(3)$ nucleation action. The new calculation is zero-temperature, flat-space $O(4)$, with gravity omitted. A finite-temperature extension requires the resummed effective potential, the thermally active mediator content, possible friction and the maximum post-inflation temperature. Inferring thermal bubble rates by dividing the present $O(4)$ action by temperature would be incorrect. Likewise, reinstating a stable-wall calculation requires a candidate whose CP-breaking vacua actually lie on the global branch.

The eighteen Python tests independently integrate the compact shell with numerical quadrature, check rational interpolation, endpoint smoothness, action-scaling stationarity and the negative-action escape, reject invalid cores and exterior energies, prove scale cancellation on exact fixtures, and check the increased kinetic cost of extra fields. Receipt tests verify the published energy ordering, exact retained action replay, full-potential sample agreement, the 48-field lift, mediator-current matching, monotone decoupling in the retained scan, and CP degeneracy. The scientific receipt replays byte for byte in a fresh process under the specified single-thread numerical environment. These are focused mathematical and action checks; no full repository regression or new Lean theorem is claimed for this push.

The continued coordination boundary is explicit. The other flavor session owns exact tensor closure, constrained mechanisms, running, quantum corrections, physical matching and vacuum discovery. This release consumes its c9010fb conclusions and changes only the cosmology module, driver, focused tests, cosmology receipt, plotting/rendering scripts and this monograph. It commits directly to the main working branch after checking the latest head. The arithmetic and Lean streams remain intact. Future cosmology calculations can reuse the same radial compiler on a new action without borrowing protection, hierarchy or vacuum claims from incompatible branches.

The advance is a reusable decay-path compiler tied to the repository's actual published action, together with its mediator-induced kinetic geometry and exact radial witnesses. It sharpens what has to be measured next: a controlled action and least-path calculation, a dimensionful normalization and prefactor, and a viable cosmological occupation history. The conditional CP-wall and constrained-bias results from the preceding release remain useful for future globally CP-breaking candidates; they are retained below with their original hypotheses.

This advance consumes the published flavor conclusions without changing their mechanisms, matching calculations, operator algebra, scalar vacua, or cumulative flavor monograph. It adds exact rational tools for a conditional cosmological question: if the chosen flavor branch really supplies degenerate physical CP-breaking global vacua, can a proposed cosmological history avoid their walls, and can a permitted explicit bias remove them without undoing specified first-order flavor protections? No dimensionful wall tension, transition temperature, gravitational-wave

spectrum, or cosmologically viable repository benchmark is obtained in this release.

The implementation is `research/flavor_cosmology/cp_wall_tools.mjs`; its dependency-free test is `research/flavor_cosmology/test_cp_wall_tools.mjs`. Its examples and validation record are `receipts/flavor_cosmology/conditional_wall_tools.json`. Run `node research/flavor_cosmology/replay.mjs` to reproduce that receipt; `replay` retains the historical validation record and regenerates the scientific examples. Run `node research/flavor_cosmology/test_cp_wall_tools.mjs` from the repository root. These additions are isolated from the active scalar-loop and mechanism work. The exact mathematical tools were executed in the available V8 runtime; 104 assertions passed; the standalone replay reproduces the receipt byte for byte in V8. The workspace execution server failed to initialize, so neither the delivered Node command, Python, Lean, nor the full repository suite ran in this session. This difference is recorded rather than attributed to another session's tests.

The coordination snapshot is pinned to `a5d0f04aa6c8f666dfd9428747abbdf3811fe50`, published at 2026-10-06 19:04 UTC. The arithmetic stream there adds elliptic division-fibre theorems, coordinate normalization, halving transport and rational-root completeness. Its predecessor `ff5d04ab56e417bbc359bfe2c11264ac566cc4d1` supplies exact elliptic witnesses and rational division fibres. The published flavor stream is `8f18f0a037d7936f7ffee9aca1a2a2959c19190e`, published at 18:36 UTC: the protected soft-mediator completion and scalar-loop closure. The user reports that a separate flavor session now owns the full source-quartic exact proof, constrained mechanisms, coupled matching and running, leading quantum and higher-operator responses, extended vacuum tests, physical parameters, and its publication. All of those are excluded from this branch's file edits. The monitoring scope is published GitHub changes and user-supplied session updates; uncommitted work and private session execution are not observable.

The existing 48-scalar completion has a positive full local Hessian, complete finite quark masses and nonzero physical weak CP in its chosen benchmark, with stated first-order mass-phase protection. Its 251-coefficient operator census and fixed-mass response do not establish a golden depth or a predicted 66-degree phase. The earlier exact field-content CP intertwiner is an anti-linear involution on the representations that pass its consistency checks, not a class-inverting operation valid for arbitrary added representations. Those are upstream conclusions in `docs/VALENTINER_SOFT_MEDIATION_AND_LOOP_CLOSURE.md` and `docs/VALENTINER_CP_AND_GLOBAL_MODES.md`, consumed at the pinned commit. Their current numerical vacuum is not promoted here to a classified global minimum.

Let C denote a consistent CP operation on the full theory and let v be a homogeneous stationary field configuration. If the scalar potential and kinetic metric obey $V(C\phi)=V(\phi)$ and the corresponding isometry relation, Cv is stationary with the same energy and the same canonically measured fluctuation spectrum. An internal-symmetry-invariant CP-odd observable O obeys $O(g\phi)=O(\phi)$ and $O(C\phi)=-O(\phi)$. If $O(v)$ is nonzero, v and Cv cannot be in the same orbit of the internal unitary group: assuming $Cv=gv$ gives $O(v)=-O(v)$. For a quark-matched example, a nonzero Jarlskog invariant with nondegenerate spectra is such a diagnostic. It proves orbit distinction, not a count of all global minima. Distinct orbits alone do not prove disconnected vacuum components: a continuous degenerate path could join them. Isolated global minima after gauge reduction supply the additional hypothesis. A positive physical Hessian establishes local isolation; the repository's remaining global search must establish global minimality.

Counting all 1080 Valentiner matrices as independent physical vacua would therefore be wrong if an internal gauge redundancy identifies their configurations. Conversely, quotienting by ordinary internal transformations does not justify erasing spontaneous CP walls. CP reverses spatial orientation, and gauged parity/CP is not ordinary internal discrete gauge symmetry. McNamara and Reece, *Reflections on Parity Breaking*, arXiv:2212.00039, analyze exactly stable walls in the gauged-parity setting. A gravity embedding and its orientation structure are not supplied by this finite flavor representation calculation. This release leaves that embedding explicit rather than treating the phrase gauged CP as a wall-removal mechanism.

There is an exact energy result independent of any numerical minimizer. In a CP-invariant quantum theory with a compatible measure, regulator and renormalization prescription, CP-even corrections preserve equality of conjugate branch energies. Thresholds, perturbative running, higher-loop terms and CP-even higher-dimensional operators cannot alone create a CP-conjugate vacuum bias. They may change the common vacuum energy, wall profile or transition order; they do not select a sign. The thermal version holds for a CP-symmetric state, including zero CP-odd chemical potentials. A baryon asymmetry, imposed CP-odd background or explicit CP-breaking spurion changes this premise and

must be included as a physical input. Full quantum CP consistency for the repository's enlarged field content is not established by its restricted elementary intertwiner alone.

QCD also supplies no automatic loophole. At a CP-symmetric bare θ and in a CP-symmetric medium, its free energy obeys $E(\theta) = E(-\theta)$. Conjugate branches with opposite induced strong phases therefore have equal QCD energies. In the illustrative susceptibility expansion $E(\theta) = \chi \theta^2/2$, adding an independent explicit common θ_0 gives $E(\theta_0 + \theta_s) - E(\theta_0 - \theta_s) = 2 \chi \theta_0 \theta_s$. That bias relies on the explicit input θ_0 and vanishes if θ_s is zero. It is not a mechanism that follows from protected Nelson–Barr matching. Asadi, Homiller, Lu and Reece, Chiral Nelson–Barr Models: Quality and Cosmology, arXiv:2212.03882, discuss the interaction between strong-CP quality, inflation and reheating in concrete models.

The constructive wall result begins with two supplied degenerate global minima and a continuously differentiable path $\gamma(t)$, $0 \leq t \leq 1$, in canonically normalized real fields. Write $W(t) = V(\gamma(t)) - V_{\min}$ and $K(t) =$ the Euclidean squared norm of $\gamma'(t)$. Provided W is nonnegative, minimization of the one-dimensional spatial parametrization of that fixed path gives a candidate interface tension equal to the integral of $\sqrt{2WK}$. The true infimum of the interface tension cannot exceed the value from this path. Cauchy–Schwarz then gives the useful exact squared bound

$$\sigma_{\min}^2 \leq \sigma_{\text{path}}^2 \leq 2 \left(\int_0^1 W(t) dt \right) \left(\int_0^1 K(t) dt \right).$$

One sees the first step by introducing $s = dt/dz > 0$ in the static energy: its density integrated over t is $K s/2 + W/s$, minimized pointwise at $s = \sqrt{2W/K}$. Endpoint singularities or interior zeros are understood by limiting finite-energy profiles. Additional paths can lower the upper bound; no lower bound follows. The argument assumes the declared degenerate global minima. Metastable minima with a deeper third branch do not by themselves define a stable cosmological wall.

For rational polynomial inputs, the code computes both integrals exactly with BigInt fractions. It checks that W and its derivative vanish at both endpoints, divides W by t squared times $(1-t)$ squared, and checks nonnegative Bernstein coefficients of the residual polynomial. Nonnegative Bernstein coefficients certify nonnegativity on the unit interval; failure is deliberately inconclusive. This is not a complete polynomial positivity algorithm. The code accepts a caller-supplied W , so it does not independently verify that W is the composition of a multivariate repository potential with γ . Nor does it certify the endpoints' global minimality. Those hypotheses are written into each receipt. Algebraic endpoints and a nonconstant field metric require further exact-field or interval adapters.

For the canonical one-field benchmark $V(a) = (a^2 - 1)^2/4$ and $\gamma(t) = 2t - 1$, $W = 4t^2(1-t)^2$ and $K = 4$. Their integrals are $2/15$ and 4 . The exact upper bound is $\sigma^2 \leq 16/15$. The known kink has $\sigma = 2\sqrt{2}/3$, hence $\sigma^2 = 8/9$; this gives an independent strict check on the variational bound. These are dimensionless toy data. No repository tension has been assigned this value. With general positive λ and v , the same canonical quartic has tension $2\sqrt{2}\lambda v^3/3$, illustrating why a dimensionless flavor Hessian cannot fix an astrophysical wall tension without a physical normalization and scale.

An explicit bias has a separate, precisely bounded algebraic question. Let O_i be specified CP-odd operators and c_i their small real coefficients. To first order, the relaxed branch-energy difference is $b^T c$, where $b_i = 2 O_i(v)$, up to the chosen sign of which branch is subtracted. The derivative of a relaxed energy is the operator value at its unperturbed stationary point; the first-order vacuum shift does not contribute because the unperturbed potential gradient vanishes. The vacuum shift itself is $-H^{-1}$ times the summed c_i gradient O_i . A protected observable Q has first-order response equal to its direct coefficient response minus gradient $Q^T H^{-1}$ gradient O_i . Physical matching must provide that direct response and the properly gauge-reduced Hessian; the present module does not calculate either.

Suppose rows of A collect responses required to vanish to first order, for example a supplied strong-CP response and specified CKM responses. Choose an explicit positive definite coefficient metric R and a budget $c^T R c \leq \rho^2$. The mathematical problem is to maximize $|b^T c|$ subject to $A c = 0$. Discard redundant rows of A by exact elimination. Set $x = R^{-1} b$ and

$$z = R^{-1} b - R^{-1} A^T (A R^{-1} A^T)^{-1} A R^{-1} b.$$

Then $Az=0$, $z^T R z = b^T z \geq 0$, and the exact maximum squared bias is $\rho^2 b^T z$. For a nonzero result, the maximizing coefficient vector is $\rho z / \sqrt{b^T z}$. This follows by taking the R -orthogonal projection of x onto $\ker A$ and applying Cauchy–Schwarz in the R metric. The module gives the rational projected direction, exact squared support and all constraint residuals. A vanishing support is a first-order impossibility result under the supplied constraints and coefficient budget, not an all-orders theorem that a cosmology is impossible.

The exact criterion has a useful sharp interpretation. Because R is positive definite, no allowed first-order bias exists exactly when b lies in the row span of A . If a CP-odd operator distinguishes the conjugate vacua but its bias functional is fully tied to the responses being protected, forbidding those responses also forbids that bias. If b has a surviving component in the constrained subspace, an algebraic first-order direction exists. Its physical size, radiative stability, higher-order strong phase, naturalness and full thermal dynamics remain upstream model calculations. Treating a cancellation $A c=0$ as a symmetry-derived Wilson-coefficient relation would exceed this result.

The hand-worked example uses $R=\text{diag}(1,2,3)$, $b=(2,2,2)$, and the response constraint $c_1+c_2=0$. Its projected direction is $(0,0,2/3)$, so the unit-budget squared maximum is $4/3$, versus $22/3$ without the constraint. A duplicate constraint $2c_1+2c_2=0$ changes nothing. In a two-operator example with $R=I$ and $b=(2,2)$, the same response constraint removes all bias: the support is exactly zero. Twenty-four additional positive definite rational examples check that the projected direction satisfies every original constraint, that its R norm equals its bias pairing, and that adding constraints cannot enlarge the support. No rows in these examples are claimed to be the actual repository strong-CP derivative.

The thermal adapter is deliberately a transparent toy. For a single real CP-odd field with $V(a,T)=\lambda(a^2-v^2)^2/4+cT^2$, positive λ , c and v^2 give $T_c^2=\lambda v^2/c$. Below T_c the conjugate minima have $a^2=v^2-cT^2/\lambda$ and remain exactly energy-degenerate. The example $\lambda=2$, $v^2=9$, $c=3$ gives $T_c^2=6$. The actual 48-scalar system has independent portals and mixed thermal masses. Its critical temperature, possible symmetry nonrestoration, transition order, tunneling and nonthermal restoration are open. In a thermal restoration scenario, maximum temperature after inflation matters; the final reheating temperature alone can miss an earlier hotter epoch. Preheating and inflationary fluctuations likewise need their own calculation.

For bookkeeping only, `classicalScalingScreen` records an explicitly declared horizon-curvature estimate: $\rho_{\text{wall}}=\text{area}\cdot\sigma\cdot H$ and $H_{\text{ann}}=\Delta V/(\text{annihilation}\cdot\sigma)$, with reduced Planck mass M . Equating wall energy to $3 M^2 H^2$ gives $H_{\text{dom}}=\text{area}\cdot\sigma/(3 M^2)$. In this convention, annihilation before both a supplied deadline H_d and wall domination asks for ΔV greater than $\text{annihilation}\cdot\sigma\cdot\max(H_d, H_{\text{dom}})$. All quantities, area and annihilation factors are caller inputs. This is a phenomenological screen, not a rigorous cosmological bound or a simulation.

Combining that declared screen with the exact coefficient envelope yields a conditional test: its required ΔV^2 must be below $\rho^2 b^T z$ to have any linearized coefficient direction within the chosen budget. A certified tension upper bound can supply a conservative sufficient threshold within this screen because both threshold terms increase with σ . An upper bound that fails the test cannot establish impossibility: the true tension might be smaller. Likewise, a support zero under homogeneous first-order response constraints does not exclude quadratic-order bias, inflationary removal or a different field content. Classical estimates are not promoted to universal dynamics. Babichev and collaborators, arXiv:2504.07902, find different bias dependence in their simulated real-scalar model, reinforcing the need to calibrate a physical network before claiming a gravitational-wave prediction.

The bounded handoff to the active mechanism session is now concrete. It can supply a globally classified pair, a dimensionful canonical metric and potential, physical CP consistency for the complete content, a set of allowed CP-odd operators if explicit breaking is admitted, and their direct plus relaxed strong-CP/CKM response rows. The separate cosmology layer can then consume these inputs to compute path tension bounds, constrained bias support and thermal-history conditions without editing its loop algebra or redoing its matching. If exact CP must remain fundamental, the explicit-bias route is unavailable and the remaining cosmological questions are inflationary removal, restoration avoidance and the actual spacetime realization of CP. These are distinct from choosing a locally stable flavor frame.

The contribution is the executable rational interface between supplied flavor protection constraints and cosmological bias, together with a conservative polynomial-path tension bound and a recorded concurrency boundary. CP-domain-wall existence, their standard cosmological problem, variational wall mechanics and constrained quadratic optimization are established mathematics and physics. No claim of a new fundamental wall mechanism, a new CKM prediction or publication priority is made. References are McNamara and Reece, <https://arxiv.org/abs/2212.00039>; Asadi, Homiller, Lu and Reece, <https://arxiv.org/abs/2212.03882>; Saikawa, <https://arxiv.org/abs/1703.02576>; and Babichev et al., <https://arxiv.org/abs/2504.07902>.